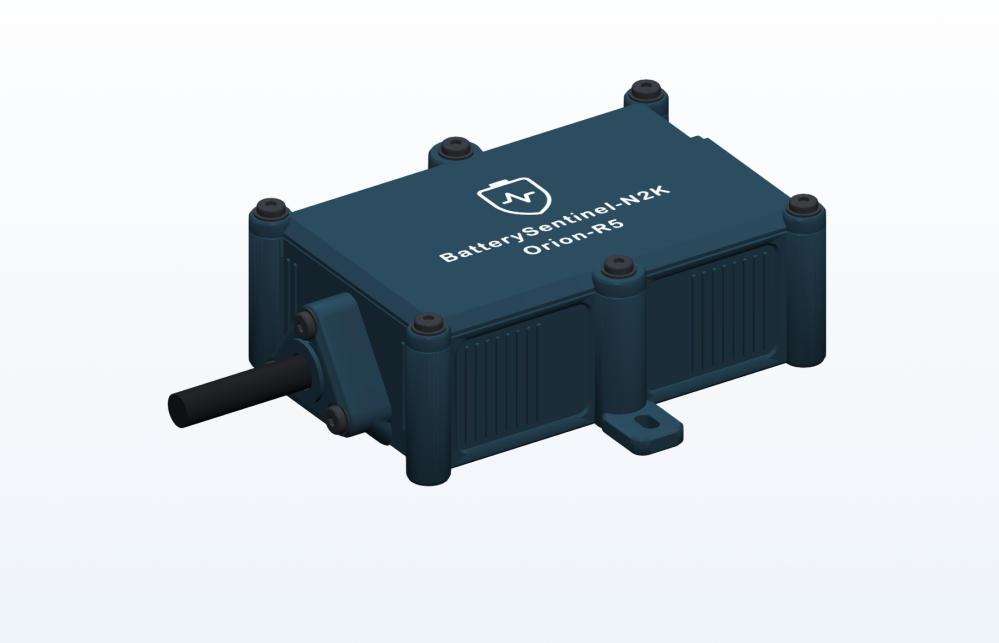
# Manufacturing and TPU Air

V2.3 / P2S / 0.4 MM NOZZLE / CALIBRATION FIRST




Actual V2.3 CAD: thin walls, projecting screw bosses, shallow rib fields and flush artwork. Screws and cable are reference hardware.

Material: **SainSmart Flex TPU Air / lightweight foaming TPU, advertised 77A-93A**. PETG carries the structural loads. The seal and clamp are independent. The file set includes production parts, nine alternative sleeves, fit fixtures, a flow coupon and a separate hardness comparison pad.

**Target hardness is not a temperature promise**

The Shore ranges in this manual are engineering trial targets. They are not measured results or a supplier-certified temperature-to-hardness table. Accept a setting only after dimensional, compression, leak and retention checks. Do not transfer Siraya Tech TPU Air flow/hardness tables to SainSmart.

Files are prepared for reslicing after calibration, not direct printing of unverified cached G-code. Preserve the prior project and its user test plates. Select the real nozzle and build plate before every print; nozzle type changes require a new calibration.

# Hardness targets by function

01 / USE MODERATE FOAMING FIRST

| Part | Target Shore A | First temperature / comparison |
|---|---|---|
| Lid gasket | 80-85 A | 245 C; compare 235 and 255 C |
| USB gasket | 83-88 A | 235 C; compare 245 C |
| Switch gasket | 83-88 A | 235 C; compare 245 C |
| Cable sealing sleeve | 85-90 A | 235 C; compare 245 C |
| Optional clamp liner | Dense 95A; unfoamed | Use its own manufacturer profile |
| Cap / clamp / case | PETG; no Shore A target | Use the supplied PETG process with calibrated filament |

The larger lid loop benefits from the lowest contact force. Small service seals need enough body to keep their continuous bead shape. The cable sleeve is slightly firmer to retain accurate bores and shoulder contact; it does not provide strain relief. These ranges are a starting design preference, not a validated optimum.

Do not aim for 77A everywhere. Maximum foam may lose dimensional accuracy, recovery or impermeability. If a 90A sample passes all functional tests while a softer sample seeps, choose the passing sample and record the departure from the target. Conversely, an in-range Shore value does not prove a watertight seal.

SainSmart supplies a 210-270 C nozzle range, 40-60 C bed range and moderate-speed guidance. The relationship between temperature and final hardness depends on speed, nozzle residence time, flow, cooling and print structure. The comparison temperatures here are experiments within that range, not certified hardness settings. [1]

# TPU Air starting process

02 / SETTINGS SHARED BY THE TEMPERATURE TRIALS

| Setting | Initial value / qualification |
|---|---|
| Nozzle / layer / line width | 0.4 mm / 0.20 mm / nominal 0.42 mm; recalibrate for actual nozzle |
| Temperature jobs | 235, 245 and 255 C as separate profiles/jobs |
| First layer / other nozzle temperature | Same temperature; stabilise before printing |
| Bed | 50 C on the selected compatible plate |
| Walls / infill / top-bottom | 3 walls / 100% / 5 and 5 layers |
| Wall + solid infill speed | 20 mm/s; first layer 15 mm/s |
| Maximum volumetric speed | 2.0 mm3/s ceiling; do not use PETG HF limits |
| Cooling | 30% after initial adhesion; auxiliary fan off |
| Retraction | 0.4 mm at 20 mm/s as a trial; tune for actual feed |
| Flow ratio | 1.00 is an UNCALIBRATED seed; calibrate before production |
| Feed / drying | External spool, low drag; spool-specific drying instructions |
| Surface | No ironing; no support scars on functional seal faces |

100% infill removes intentionally sparse voids; individual extrusions can still foam. It neither prevents foaming nor guarantees a watertight surface. Avoid low infill or deliberate gaps as a softness control. There is no proven unfoamed skin in a uniformly hot printed seal.

Keep speed and cooling stable while comparing temperatures. Small-layer slowdown and the first-layer speed can change expansion. The supplied trials disable cooling-driven slowdown; inspect the actual preview and test the small service seals at their real speed. Stop a print if feed buckles or repeated underextrusion appears.

# Temperature and flow calibration

03 / COMPLETE THIS BEFORE FULL SEAL PRINTS

### 1 Prepare dry material

Check the spool label and batch. Dry with this material/spool supplier instruction and feed from dry storage. Moisture bubbling is not controlled foaming; do not mistake it for a softer successful print.

### 2 Establish a baseline

Print flow_wall_coupon at 235 C, 20 mm/s, 0.20 mm layers and seed flow 1.00. This small coupon contains a nominal 0.9 mm wall. Check the slicer actually generates continuous paths at the intended wall width.

### 3 Correct expansion

Measure several wall positions and inspect the solid regions. If overexpanded, reduce flow progressively. Old flow x target/measured width is a rough first correction only; confirm again. Reject settings with gaps, swollen seams or unstable feed.

### 4 Repeat at each temperature

Make independent 245 C and 255 C jobs and calibrate flow separately. Record nozzle type, speed, flow and dimensions. Never copy a 50% flow guess or assume one flow ratio fits all temperatures.

### 5 Measure a comparison pad

With each calibrated setting, print shore_comparison_pad and the face/cable specimens. Use the procedure on the next page. The pad and thin seals can differ in thermal history; qualify the actual seal too.

### 6 Select by function

For each seal choose the least aggressive foaming that reaches suitable compliance and passes the contact/leak/recovery tests. If targets are missed, adjust in 5 C steps within the supplier range and recalibrate flow. Treat 260-270 C as a later experiment, not the default.

# Measuring Shore A

04 / A COMPARISON, NOT A CERTIFIED MATERIAL TEST

Use the supplied **50 x 50 x 6 mm** flat comparison pad, printed with 100% infill and the same material/process as the trial. Bareiss describes flat specimens of at least 6 mm thickness for its Shore instruments. Follow your durometer manual for the presser foot, load, dwell time and test-point spacing. [2]

Allow all samples the same conditioning time and temperature; a practical internal comparison is after 24 hours in the same room. Support the pad flat on a rigid surface. Use a stand if available, approach perpendicular and avoid an edge, embossed mark, seam or visibly defective area. Never measure the 1.8 mm gasket against a hard table and call that the bulk Shore value.

Take at least five spaced readings using the instrument-prescribed dwell. Record all readings, the mean, range, dwell, room temperature and conditioning. Apply the same method throughout. If instrument instructions allow both, record separate 1-second and 15-second values; do not mix them in one average. Printed foam is structured and anisotropic, so these are process-comparison values, not certified ASTM/ISO material properties.

| Trial record | Write measured values |
|---|---|
| Temperature / nozzle type / speed | _____ |
| Flow / fan / batch / drying | _____ |
| Pad size / mass / conditioning | _____ |
| Readings 1 / 2 / 3 / 4 / 5 | _____ |
| Mean / min-max / dwell / ambient | _____ |
| Actual gasket contact / leak / recovery | _____ |

**Without a durometer**

Record hardness as unmeasured. A finger squeeze or scale fixture can compare force and recovery, but cannot establish Shore A. You can still screen the functional seals; do not label them with a fabricated Shore value.

# Print orientation and rigid parts

05 / SEPARATE PETG AND TPU JOBS

| Part | Orientation / critical point |
|---|---|
| Base + entry coupon | Outside floor on plate. Inspect horizontal insert/socket roofs. |
| Two-colour lid | Exterior face down. Body and markings share one transform; never arrange markings independently. |
| Service covers | Groove face down, exterior reinforcement up. Inspect the 3.4 mm groove-roof bridges before assembly. |
| Pressure cap | Flat sealing side down, reinforcement ribs up. Keep washer seats flat. |
| Upper cable clamp | Flat top down; open saddle upward, support-free. |
| Face seals | Flat carrier down; rounded bead upward. No brim touching the bead. |
| Cable sleeves | Cable axis vertical; raised shoulder bead up. No support in the bore. |
| Shims | Flat. 0.2 mm is one layer; inspect for completeness. Never mix unequal thicknesses. |

PETG: 0.20 mm layers, 6 walls, 6 top/bottom layers, 35% gyroid, outer/inner walls 40/80 mm/s, 12 mm3/s ceiling. Cap and upper clamp: 100% infill. Use compatible PETG colours. V2.3 has 3 mm free walls/floor, 0.6 mm texture, a local 6-6.6 mm sealing rim and 8 mm connector pads. Support slopes stay at or below 45 degrees from vertical. These are prototype dimensions; inspect layer bonding, flex and leaks before use.

Use the supplied print orientations and scale 100%. Choose the actual build plate and calibrated PETG/TPU filament. A smooth plate with an appropriate release layer may improve contact faces; remove parts after cooling without stretching seals. Do not automatically smooth, decimate or scale all parts to compensate for one bore.

# Fit and compression experiments

06 / SELECT A SLEEVE AND CLAMP SETTING

| Variable | Alternatives / purpose |
|---|---|
| Cable bead bore | 6.9 / 6.8 / 6.7 mm for the measured 7 mm jacket |
| Sleeve free length | 6.6 / 6.8 / 7.0 mm into a 6.0 mm seat |
| Axial squeeze | 9.1% / 11.8% / 14.3% respectively |
| Initial sleeve | b6p8_L6p8; compare adjacent alternatives if needed |
| Face seal | 1.8 mm free, 1.4 mm production stop; 22.2% nominal squeeze |
| Face test fixture | Compare 1.4 and 1.6 mm stops with a straight section cut from the spare face_test_gasket; a flat plate supplies the load |
| Clamp shims | 0.4 mm first, then 0.2 mm, then none only if needed |

The outer sleeve body has relief space. Its two internal rounded beads contact the jacket; the annular end bead contacts the pocket shoulder. Inspect both paths. A cable grip test does not establish shoulder sealing, and a watertight sleeve does not establish strain relief.

Test insert seating and cap closure on entry_fit_coupon. It reproduces the actual horizontal print orientation. Do not chase a leak by adding torque after the hard stops meet. Check flow, bead continuity, bore dimensions and jacket ovality first. Complete the final exposure test on the complete empty case, not only the open coupon.

Keep specimens compressed for 24-48 hours. Check load relaxation, cap bowing and seepage; after release, measure recovery at a fixed recorded time. Repeat under representative warmth and after opening/closing cycles. If a part takes excessive permanent set or leaks, its Shore target alone is irrelevant.

# Release checklist and sources

07 / EVIDENCE REQUIRED BEFORE BOAT INSTALLATION

## 1 Geometry and slicing

Confirm validation.json passes and package-validation.json identifies the actual slicer used. Open the prepared project, check object/material mapping, plate bounds, first-layer contact, bridges and every thin seal bead. Re-slice after any calibration change.

## 2 Material process

Record final per-temperature flow, measured pad hardness or unmeasured status, and actual seal dimensions. No supplier-approved hardness/temperature curve is supplied for this printer/nozzle.

## 3 Mechanical and water checks

Complete the assembly manual: insert fit, independent clamp retention, jacket inspection, 24-48 h recheck and empty-case leakage before/after cable flexing and service cycles.

## 4 Fallback if foam fails

If water permeates the material or it loses compression, retain the separate clamp/stiff cap arrangement and redesign the seat for a selected conventional silicone/EPDM seal or commercial cable gland. Do not claim waterproofness from a closed STL.

### Sources checked 27 September 2026

[1] SainSmart, Lightweight TPU Filament / Foaming LW-TPU 77A-93A: sainsmart.com/products/lightweight-tpu-filament. Material range and general print guidance; not a validated P2S seal recipe.

[2] Bareiss, Digital Durometer Hardness Tester: bareiss-testing.com/digital-durometer-hardness-tester/. Specimen/instrument context; follow your own instrument instructions.

Geometry and dimensions: sealed-v2/parameters.json, current PCB body model, CAD export and digital validation. Shore ranges, comparison temperatures and acceptance sequence are project engineering proposals requiring physical validation.